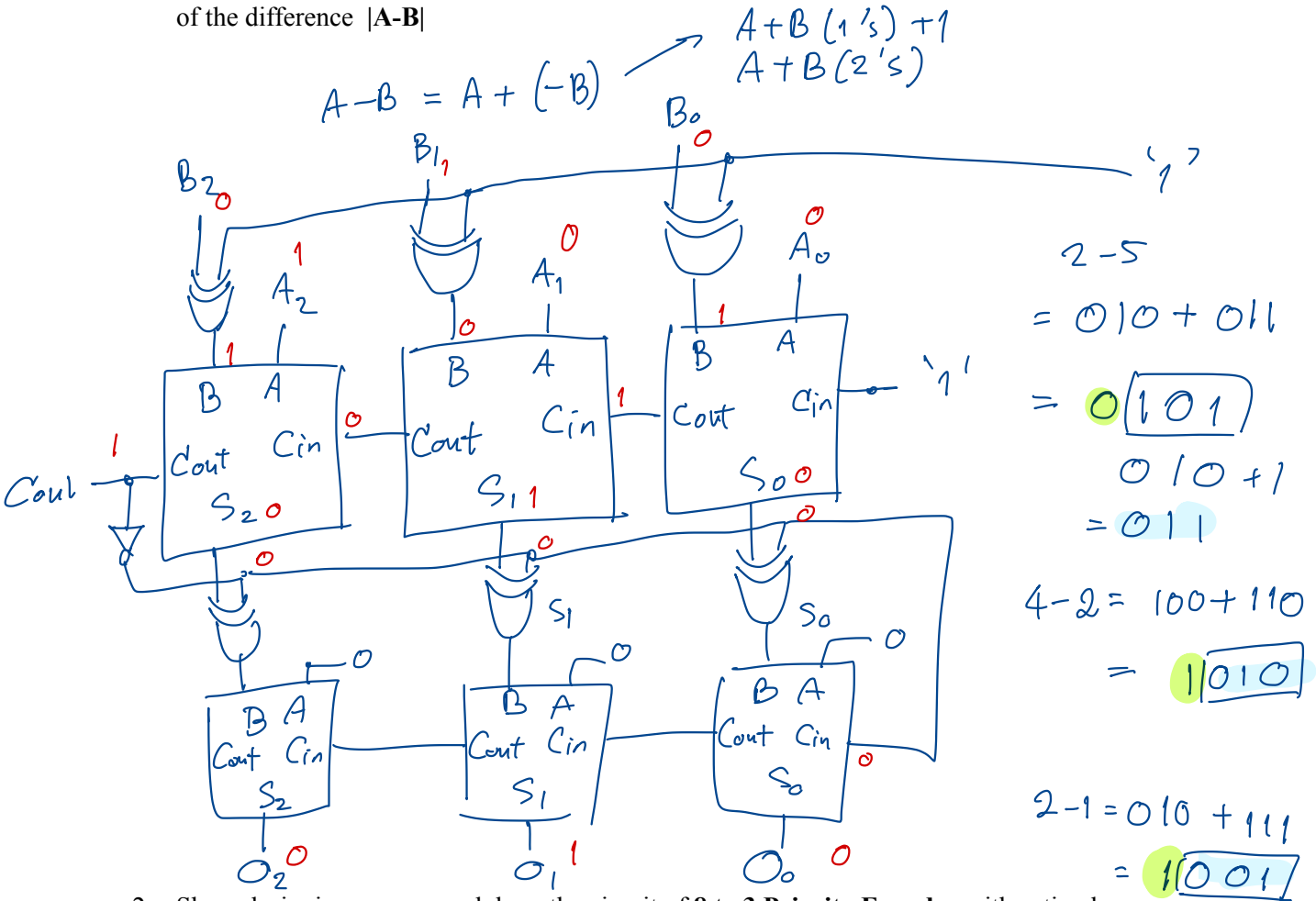
Section 1 : General and Combinational Circuit Design (10 marks)

1. Given 3 bits unsigned integer **A,B** designing a circuit that provide the output of absolute value of the difference |**A-B**|



2. Show designing process and draw the circuit of **8 to 3 Priority Encoder** with active low outputs and active high enable signal

E	I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	S_2	S_1	S_0
1	0	0	0	0	0	0	0	1	1	1	1
	0	0	0	0	0	0	1	x	1	1	0
	0	0	0	0	0	1	x	x	1	0	1
	0	0	0	0	1	x	x	x	1	0	0
	0	0	0	1	x	x	x	x	0	1	1
	0	0	1	x	x	x	x	x	0	1	0
	0	1	x	x	x	x	x	x	0	0	1
	1	x	x	x	x	x	x	x	0	0	0
0	0	0	0	0	0	0	0	0	x	x	x

$$S_0 = \bar{I}_7\bar{I}_6\bar{I}_5\bar{I}_4\bar{I}_3\bar{I}_2\bar{I}_1 I_0 + \bar{I}_7\bar{I}_6\bar{I}_5\bar{I}_4\bar{I}_3 I_2 + \bar{I}_7\bar{I}_6\bar{I}_5 I_4 \bar{I}_7 I_6$$

$$S_1 = \bar{I}_7\bar{I}_6\bar{I}_5\bar{I}_4\bar{I}_3\bar{I}_2\bar{I}_1 I_0 + \bar{I}_7\bar{I}_6\bar{I}_5\bar{I}_4\bar{I}_3 I_2 I_1 + \bar{I}_7\bar{I}_6\bar{I}_5 I_4 + \bar{I}_7\bar{I}_6 I_5$$

$$S_2 = \bar{I}_7\bar{I}_6\bar{I}_5\bar{I}_4\bar{I}_3\bar{I}_2 I_1 I_0 + \bar{I}_7\bar{I}_6\bar{I}_5 I_4\bar{I}_3\bar{I}_2 I_1 + \bar{I}_7 I_6 I_5 I_4 I_3 I_2 + \bar{I}_7\bar{I}_6\bar{I}_5\bar{I}_4 I_3$$

3. Explain difference between 3 types of shifting operation

1. Logical shift → When shifted, LSB or MSB is discarded.

   One side has to receive 0 as an external input

   0101 → shl 1010

   shr 0010

2. Circulating Shift → When shifted, LSB can be moved to MSB position or vice versa.

   cil 1011 → 0111

   cir 1001 → 1100

3. Arithmetic Shift → When right shifted, copy the sign bit as external input. This ensures that the value remains positive or negative after shifting.

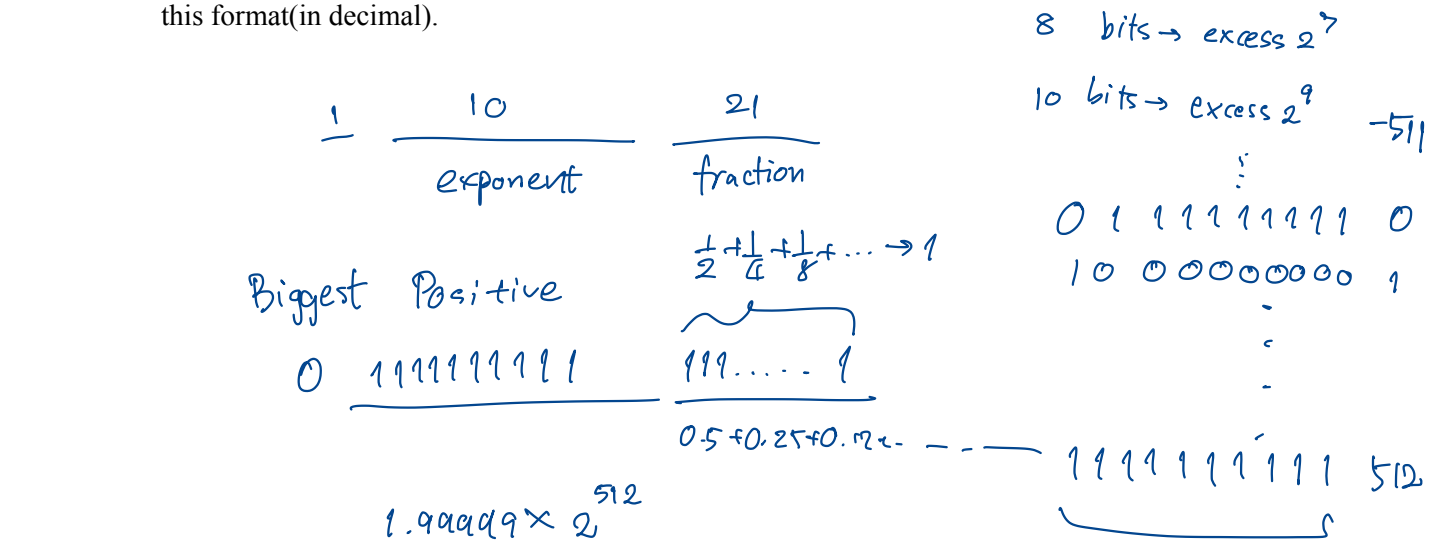
   For left shift: similar to logical

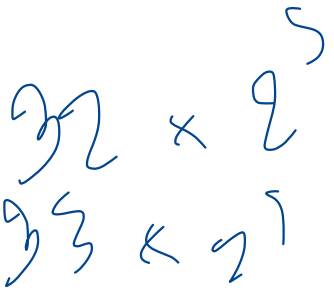
   00010 → 2
   00100 → 4
   01000 → 8 Warning
   ↳ Overflow can happen

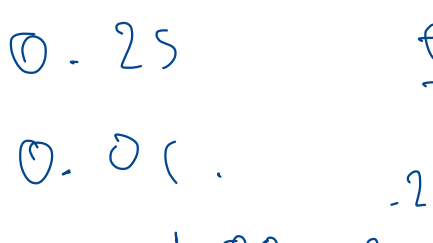
   10110 → −10
   ↓↓
   11011 → −5
   ↓↓
   11101 → −3

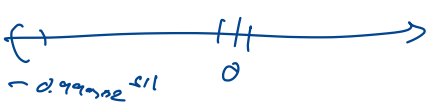
Section 2: Data representation & sequential circuit design

1. Given the floating point format (1 sign bit, 10 exponent bits, 21 fraction bits)
   What is the **biggest** positive number and **smallest** negative number that can be represented in this format(in decimal).









Smallest Negative

1 0000000000 000000.....000

$-1.000000 \times 2^{-511}$

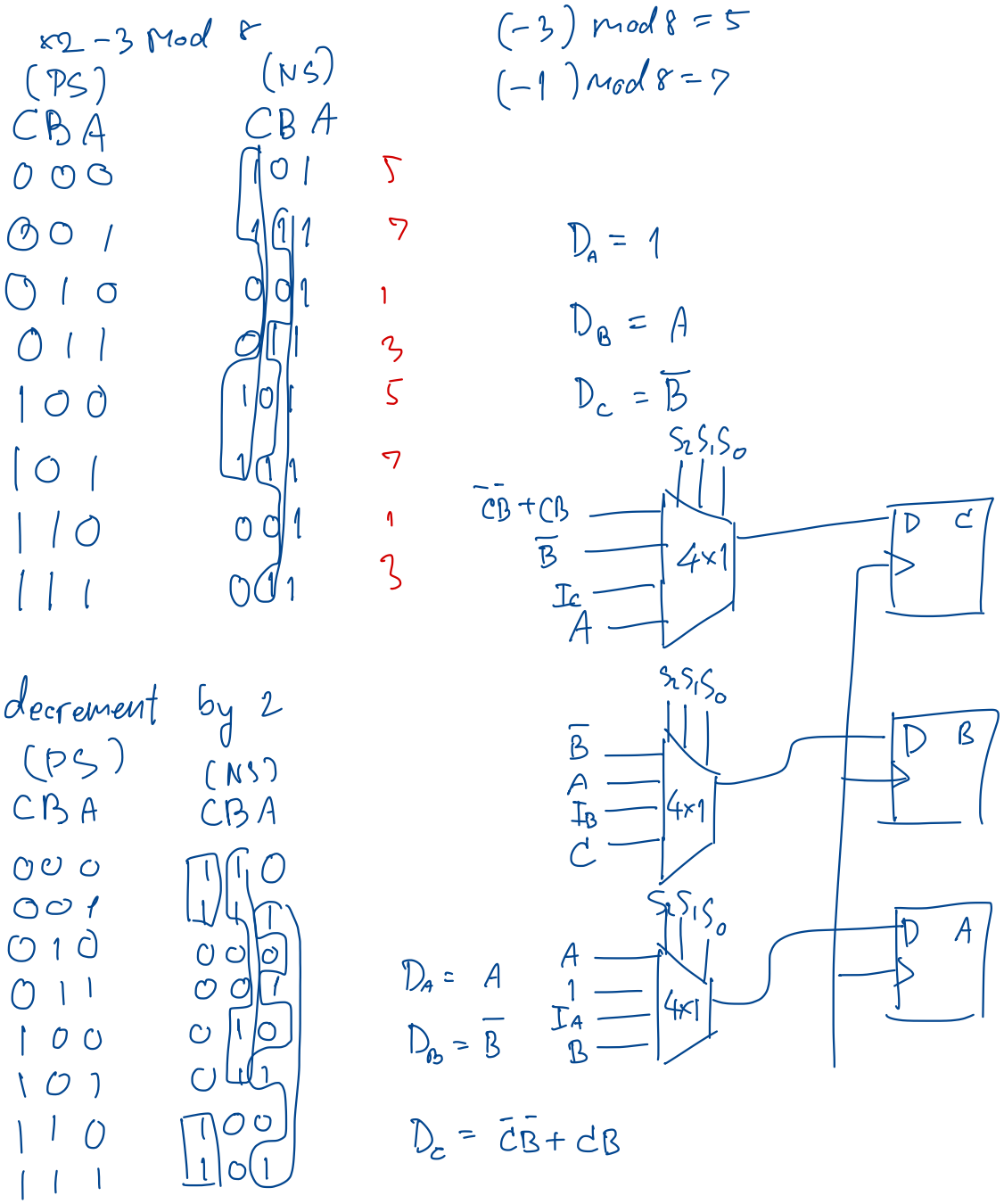


100001×2^{5}

1.00001×2^{10}

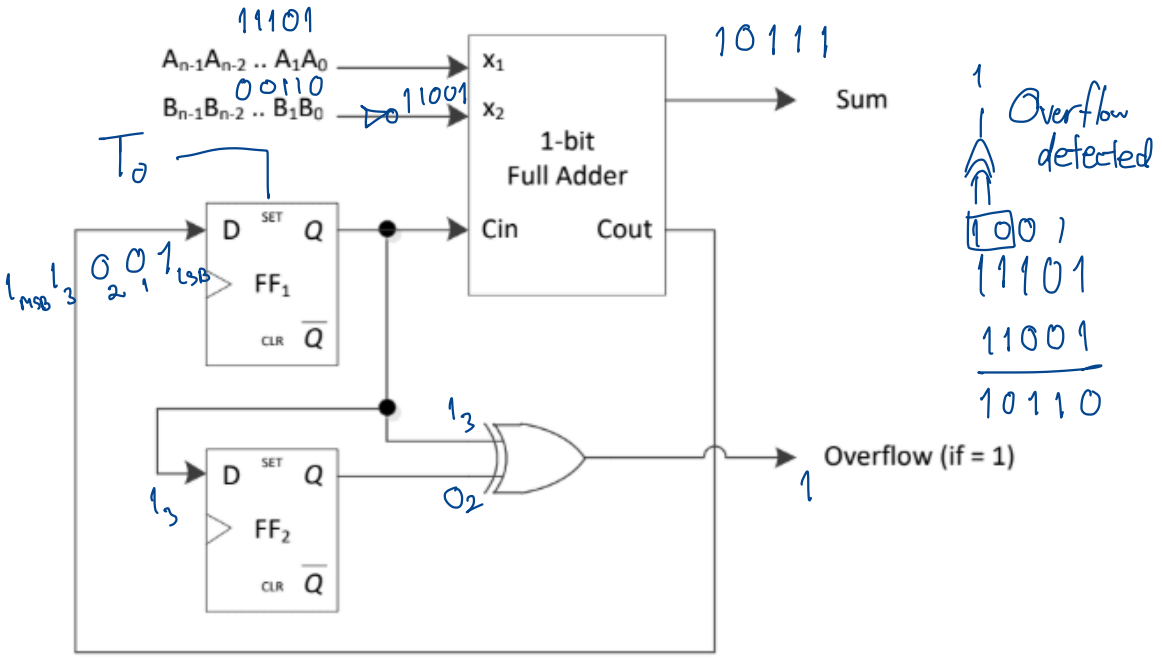
2. Design 3 bits multi-function register with this following modes

s1	s0	mode
0	0	Decrement by 2
0	1	(x2 - 3) mod 8
1	0	Parallel input
1	1	Circular shift right



Section 3: Arithmetic Circuit & Arithmetic operation

1. Explain how to modify the serial adder circuit to do subtraction operations A - B when A and B is 5 bits, A = -3 , B = 6



$A - B = A + (-B) = A + 2's\ B$

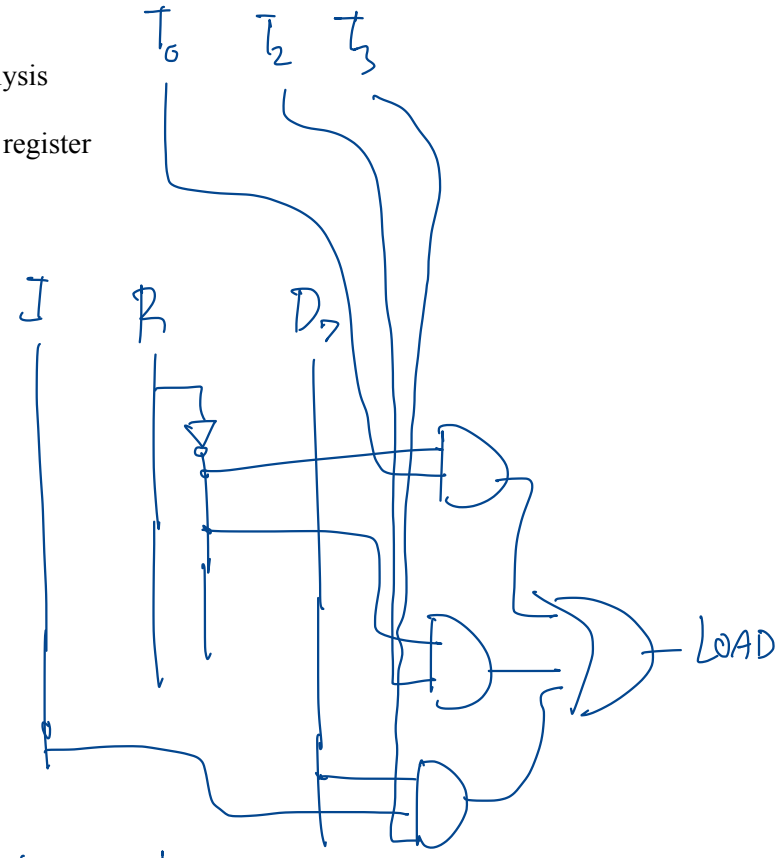
2's B means toggle and +1

First, we asynchronous set FF_1 using T_0 to give 1 to full adder. After that, we invert B. This will be equivalent to B(1's)+1 = B(2's). Therefore, we get

$$\begin{array}{r} {\scriptstyle 1\ \ 1\ \ \ \ } \\ 11101 \quad (-3) \\ + \\ 11001 \quad (6\ 1's) \\ + \\ \underline{\qquad 1 \quad (Asynchronous\ Set)} \\ 1\,10111 \quad (-9) \end{array}$$

Section4: RISC control unit design & runtime analysis

1. Design and Draw a **LOAD** control unit of register
   a. Register AR
   b. Register AC



a) AR LOAD

$R'T_0 : AR \leftarrow PC$

$R'T_2 : AR \leftarrow IR(0-11)$

$D_7'IT_3 : AR \leftarrow M[AR]$

$LOAD = R'T_0 + R'T_2 + D_7'IT_3$

b) AC LOAD , ALU

$D_0T_5 : AC \leftarrow AC \wedge DR$ AND

$D_1T_5 : AC \leftarrow AC + DR$ ADD

$D_2T_5 : AC \leftarrow DR$ DR

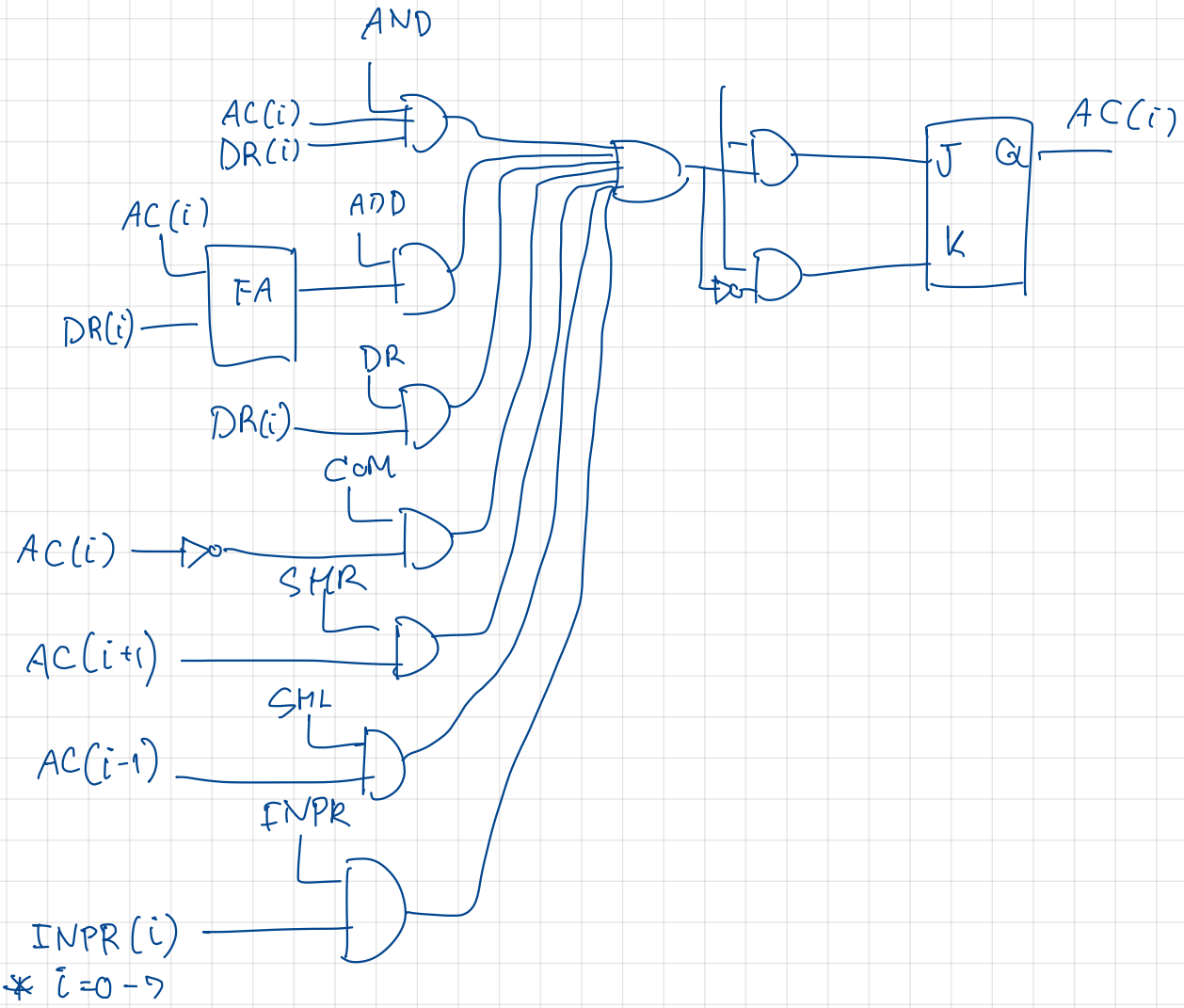
$rB_9 : AC \leftarrow \overline{AC}$ COM

$rB_7 : AC \leftarrow shr\,AC,\ AC(15) \leftarrow E$ SHR

$rB_6 : AC \leftarrow shl\,AC,\ AC(0) \leftarrow E$ SHL

$pB_{11} : AC(0-7) \leftarrow INPR$ INPR

$LOAD = D_0T_5 + D_1T_5 + D_2T_5 + rB_9 + rB_7 + rB_6 + pB_{11}$


AND
AC(i)
DR(i)
AC(i)
ADD
FA
DR(i)
DR
DR(i)
COM
AC(i)
SHR
AC(i+1)
SHL
AC(i-1)
INPR
INPR(i)
* i=0->7
J
Q
K
AC(i)

2. From this main memory, assume first line start at address 0x00

   a. Write the HEX code corresponding to content of each address

	Symbolic code	HEX code
000	LDA 00B → 200B	200B
001	STA 00C	
002	LDA 00D	
003	STA 00E	
004	CLA	
005	ADD 00C I	
006	ISZ 00C	
007	ISZ 00E	
008	BUN 005	
009	STA 00F	
00A	HLT	
00B	0000 0000 0000 0000	
00C	0000 0000 0000 0000	
00D	1111 1111 1111 0110 (F F F 6)	
00E	0000 0000 0000 0000	
	0000 0000 0001 0010	
	0000 0000 0000 0011	
	0000 0000 0000 0100	
	0000 0000 0000 0101	
	0000 0000 0000 1100	
	0000 0000 0000 0111	
	0000 0000 0000 1000	
	0000 0000 0000 1001	
	0000 0000 0000 1010	

b. Find the value of AC and flip-flop E after 40 clocks (using the code provided in the previous page)

INS	CLK	SC	AC	E	AR
LDA 00B	0	0	x	x	
	5	5	0000	x	
STA 00C	6	0	0000_H	x	
	10	4	0000_H	x	
LDA 00D	11	0	0000_H	x	
	16	5	FFF6	x	
STA 00E	17	0	FFF6	x	
	21	4	FFF6	x	
CLA	22	0	FFF6	x	
	25	3	0000_H	x	
ADD 00C I	26	0	0000_H	x	
	31	5	$200B_H$	0	
ISZ 00C	32	0	$200B_H$	0	
	38	6	$200B_H$	0	
ISZ 00E	39	0	$200B_H$	0	
	45	6	$200B_H$	0	

00C	000
00E	FFF6

00C	001
00E	FFF7

After 40 clocks
We have $200B_H$ in AC
0 in E

Practice

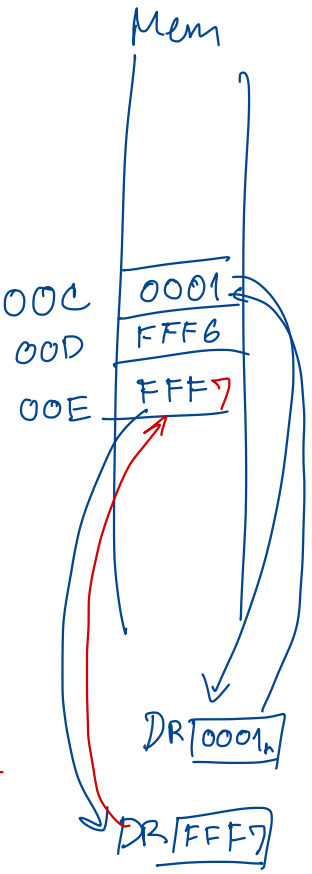
2. From this main memory, assume first line start at address 0x00

   a. Write the HEX code corresponding to content of each address

	Symbolic code	HEX code
000	LDA 00B	200B
001	STA 00C	300C
002	LDA 00D	200D
003	STA 00E	300E
004	CLA	7800
005	ADD 00C I	900C
006	ISZ 00C	600C
007	ISZ 00E	600E
008	BUN 005	4005
009	STA 00F	300F
00A	HLT	7001
00B	0000 0000 0000 0000	
00C	0000 0000 0000 0000	
00D	1111 1111 1111 0110 (F F F 6)	
00E	0000 0000 0000 0000	
00F	0000 0000 0001 0010	
010	0000 0000 0000 0011	
011	0000 0000 0000 0100	
012	0000 0000 0000 0101	
013	0000 0000 0000 1100	
014	0000 0000 0000 0111	
015	0000 0000 0000 1000	
016	0000 0000 0000 1001	
017	0000 0000 0000 1010	

b. Find the value of AC and flip-flop E after 40 clocks (using the code provided in the previous page)

INST	CLK	SC	AC	E	
LDA 00B	0	0	x	x	
	5	5	0000_H	x	PC = 001
STA 00C	6	0	0000_H	x	
	10	4	0000_H	x	PC = 002
LDA 00D	11	0	0000_H	x	
	16	5	FFF6	x	PC = 003
STA 00E	17	0	FFF6	x	
	21	4	FFF6	x	PC = 004
CLA	22	0	FFF6	x	
	25	3	0000	x	PC = 005
ADD 00CI	26	0	0000	x	
	31	5	$200B_H$	0	PC = 006
ISZ 00C	32	0	$200B_H$	0	
	38	6	$200B_H$	0	PC = 007
ISZ 00E	39	0	$200B_H$	0	40 clocks is around here
	45	6	$200B_H$	0	



After 40 clocks

$AC = 200B_H$

$E = 0$

10:48 Fri 31 Oct
96%
3
CA midterm | Midterm_Moc... | CA Midterm | CA midterm_p... | 3

Instructor: Thanadol PRITRANAN Total Marks: 50

Date: 11 ตุลาคม 2564 Hours: 1300 - 1440

Instruction: Answer all questions and submit to our Google Classroom

Outline of Midterm Examination:

Questions 1 & 2/ 25 minutes/ 12 & 10 marks (1325-1350) Control Circuit Design

- 5 min break –

Questions 3/ 20 minutes/ 16 marks (1355-1415) Runtime Analysis

- 5 min break –

Questions 4/ 20 minutes/ 12 marks (1420-1440) Interrupt Mechanism

(16 คะแนน) Runtime Analysis

(16 คะแนน) **Q3.** กำหนดให้ x คือค่า 3 digits หลังของรหัสนักศึกษา mod ด้วย 5 (เช่น นักศึกษาที่มีรหัสนักศึกษาลงท้ายด้วย 456 ค่า x จะเท่ากับ 456 mod 5 .. x = 1) จากข้อมูลที่กำหนดให้ สมมติขั้นตอนการ fetch คำสั่ง CLE จาก address 100H เริ่มที่ t_m, ค่าใน register AC เมื่อสิ้น clock t_{m+35+x} จะเท่ากับเท่าไหร่ และให้อธิบายสั้นๆ ถึงแนวทางการหาคำตอบ

$x = (117) \bmod 5$

$= 2$

$t_{m+35+x} = t_{m+37}$

Main Memory

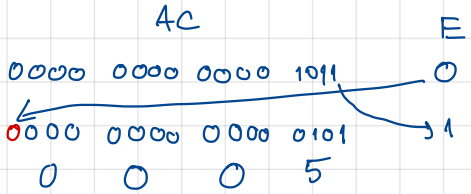
100_H	CLE	t_m
101	LDA 113_H	
102	CIR	
103	STA 113_H	
104	SZE	
105	BUN 107_H	
106	BUN $10B_H$	
107	LDA 112_H	
108	ADD 114_H	
109	STA 114_H	
10A	CLE	
10B	LDA 112_H	
10C	CIL	
10D	STA 112_H	
10E	ISZ 111_H	
10F	BUN 111_H	
110	HLT	
111	$FFF8_H$	
112	$000F_H$	
113	$000B_H$	
114	0000_H	

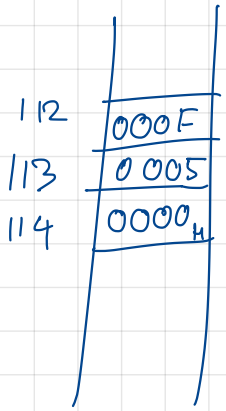
Main Memory

100_H	CLE
	LDA 113_H
	CIR

1 of 3

Inst	Clk	SC	AC	PC	E
CLE	t_m	0	X	100	X
	t_{m+3}	3	X	101	0
LDA 113	t_{m+4}	0	X	101	0
	t_{m+9}	5	$000B_H$	102	0
CIR	t_{m+10}	0	$000B_H$	102	0
	t_{m+13}	3	0005_H	103	1
STA 113	t_{m+14}	0	0005_H	103	1
	t_{m+18}	4	0005_H	104	1
SZE	t_{m+19}	0	0005_H	104	1
	t_{m+22}	3	0005_H	105	1
BUN 107	t_{m+23}	0	0005_H	105	1
	t_{m+27}	4	0005_H	107	1
LDA 112	t_{m+28}	0	0005_H	107	1
	t_{m+33}	5	$000F_H$	108	1
ADD 114	t_{m+34}	0	$000F_H$	108	1
	t_{m+39}	5	$000F_H$	109	0





0000_H
$000F$
$000F$

At t_{m+37}, AC will have the value of $000F_H$

34-35 Fetch
T_0 T_1
$T_2 \rightarrow 36$
$T_3 \rightarrow 37$